

Database Benchmark Report

Automatically generated benchmark report

Date: 2026-07-02 11:28:26

Operating System: Linux

Python: 3.12.3

Databases: H2, HSQLDB

Benchmarks: 8

Measured Runs: 20

Warm-up Runs: 5

Benchmark Summary

Number of databases: **2**

Benchmark categories: **8**

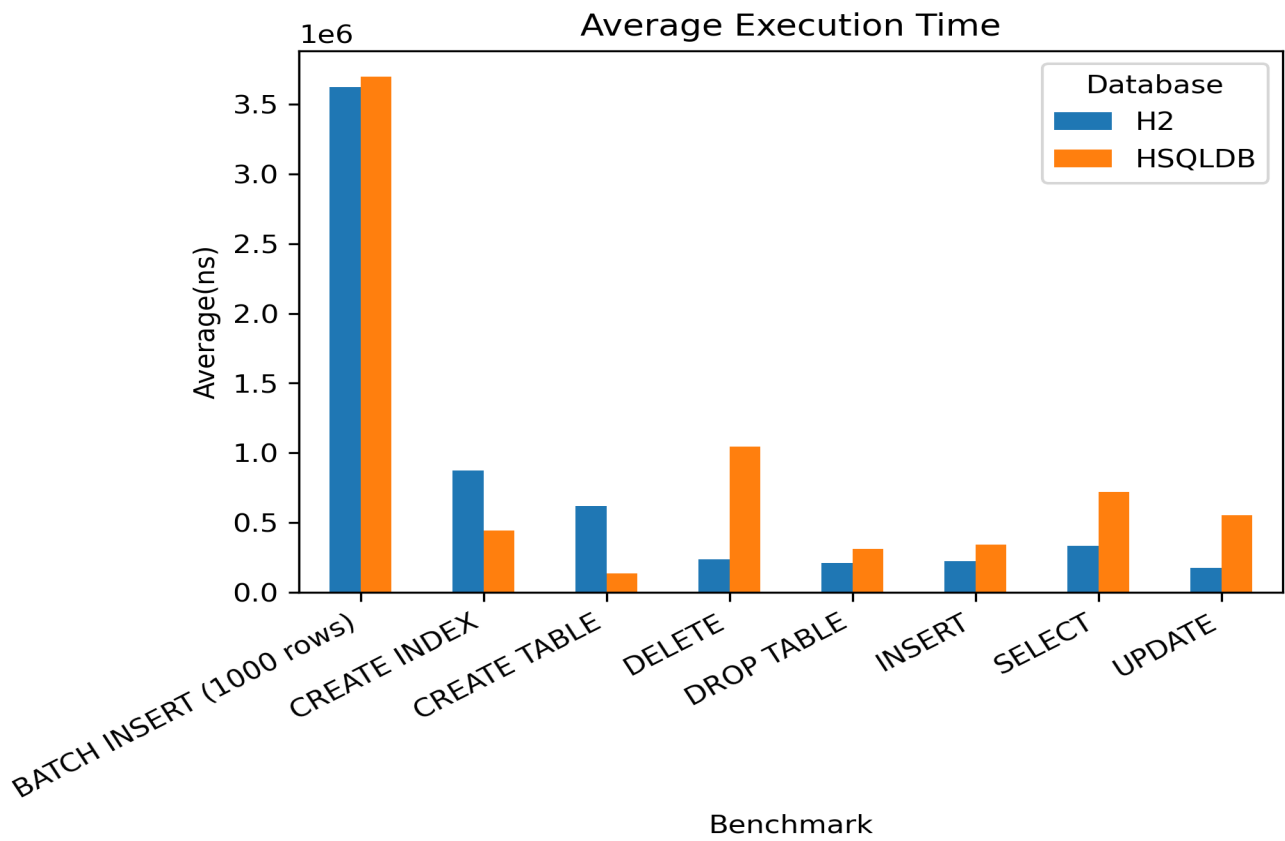
Benchmark results: **16**

This report summarizes execution-time measurements for embedded relational database management systems. All measurements were performed after JVM warm-up iterations and repeated multiple times to improve statistical reliability.

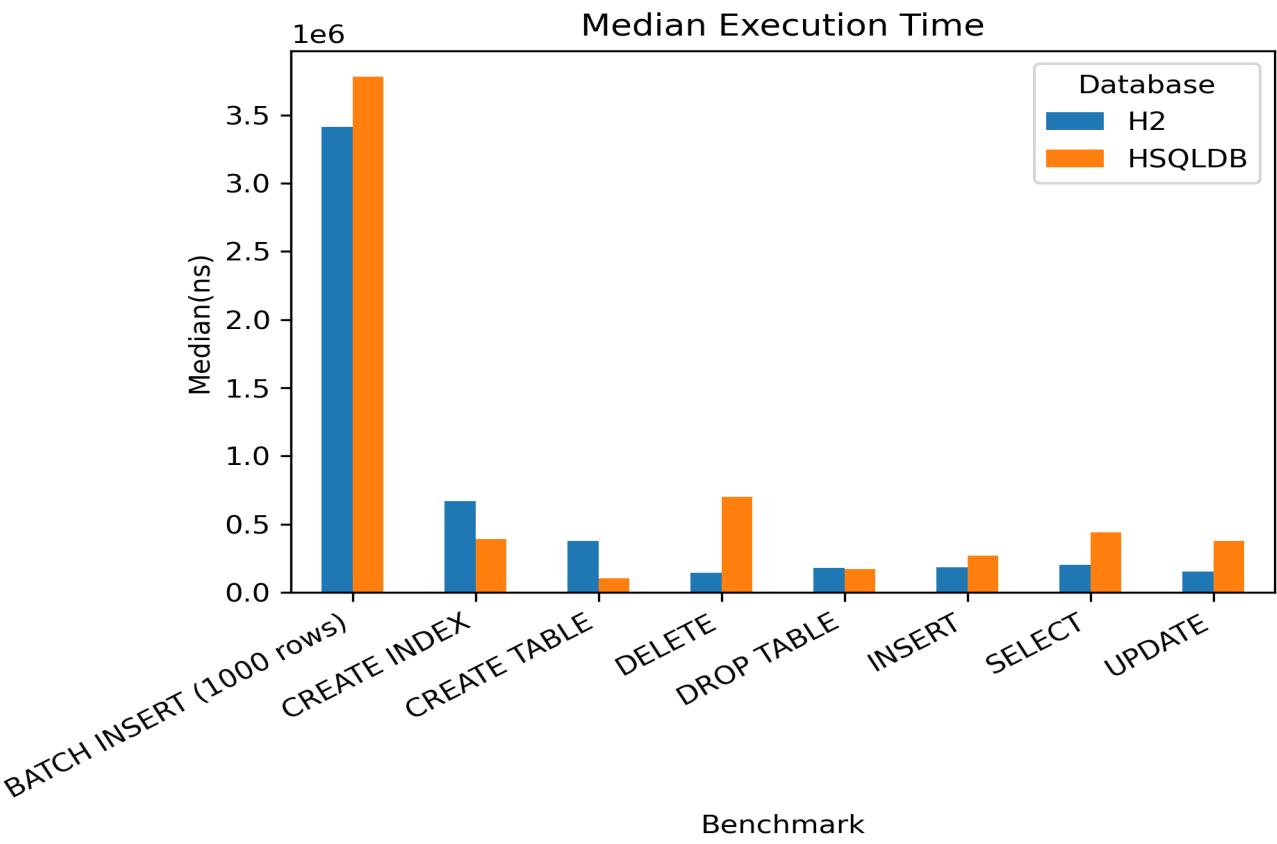
Summary Statistics

Database	Benchmark	Average	Median	Std Dev	Minimum	Maximum
H2	CREATE TABLE	621681	375501	563646	250939	2313670
H2	INSERT	225121	183157	126780	142176	703784
H2	BATCH INSERT (1000 rows)	3621863	3410333	1915878	2000295	11611660
H2	SELECT	335075	202428	495391	155791	2466816
H2	UPDATE	173912	154439	38905	142587	265516
H2	DELETE	238105	144830	246343	130884	1218596
H2	DROP TABLE	212929	178313	119342	150050	703463
H2	CREATE INDEX	872257	670532	543296	545439	2514164
HSQldb	CREATE TABLE	137618	103228	124754	86181	670932
HSQldb	INSERT	343002	268547	162994	216374	736044
HSQldb	BATCH INSERT (1000 rows)	3696618	3780274	1121656	1646104	5665458
HSQldb	SELECT	720222	439090	541041	293137	2309773
HSQldb	UPDATE	553895	376178	422690	262350	1972904
HSQldb	DELETE	1046947	702582	768832	254355	2827880
HSQldb	DROP TABLE	311148	172422	353850	143678	1576874
HSQldb	CREATE INDEX	443077	389849	156665	320499	1003043

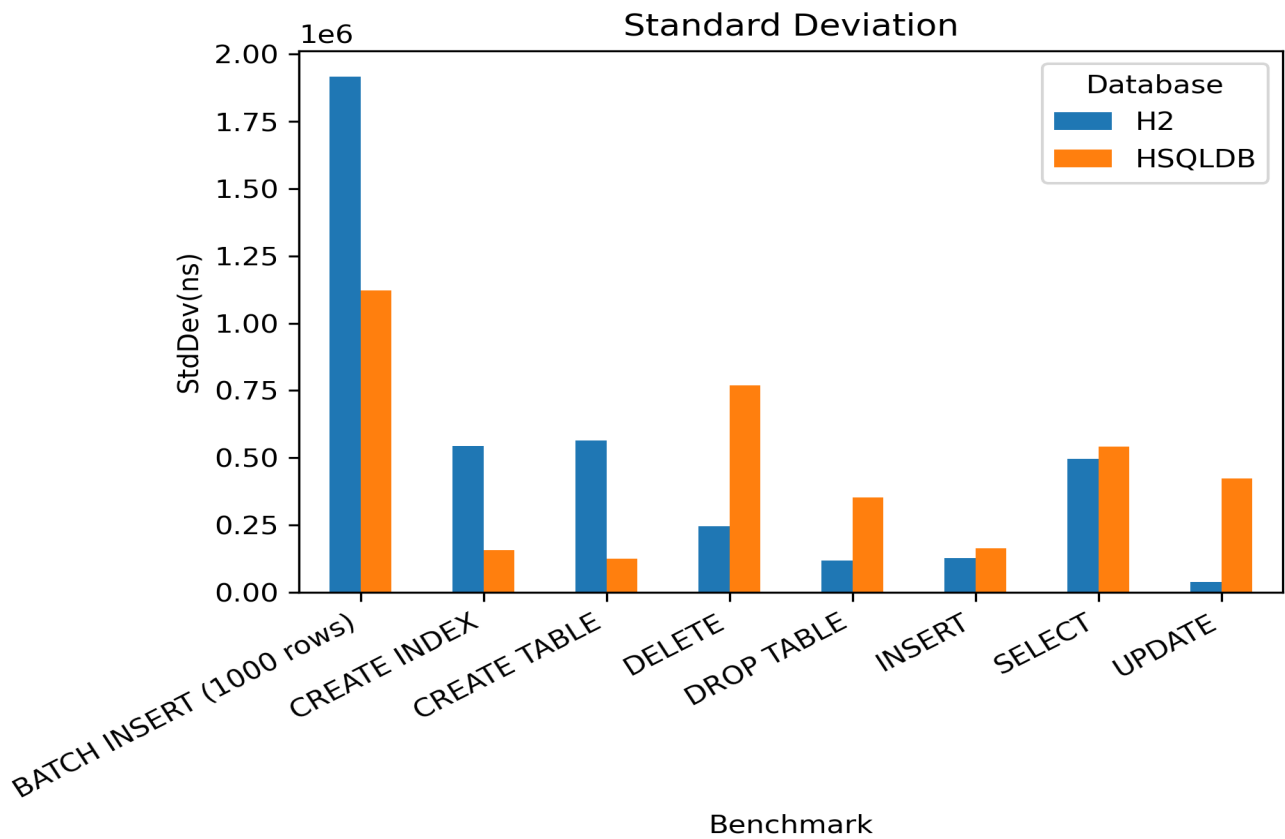
Average Execution Time



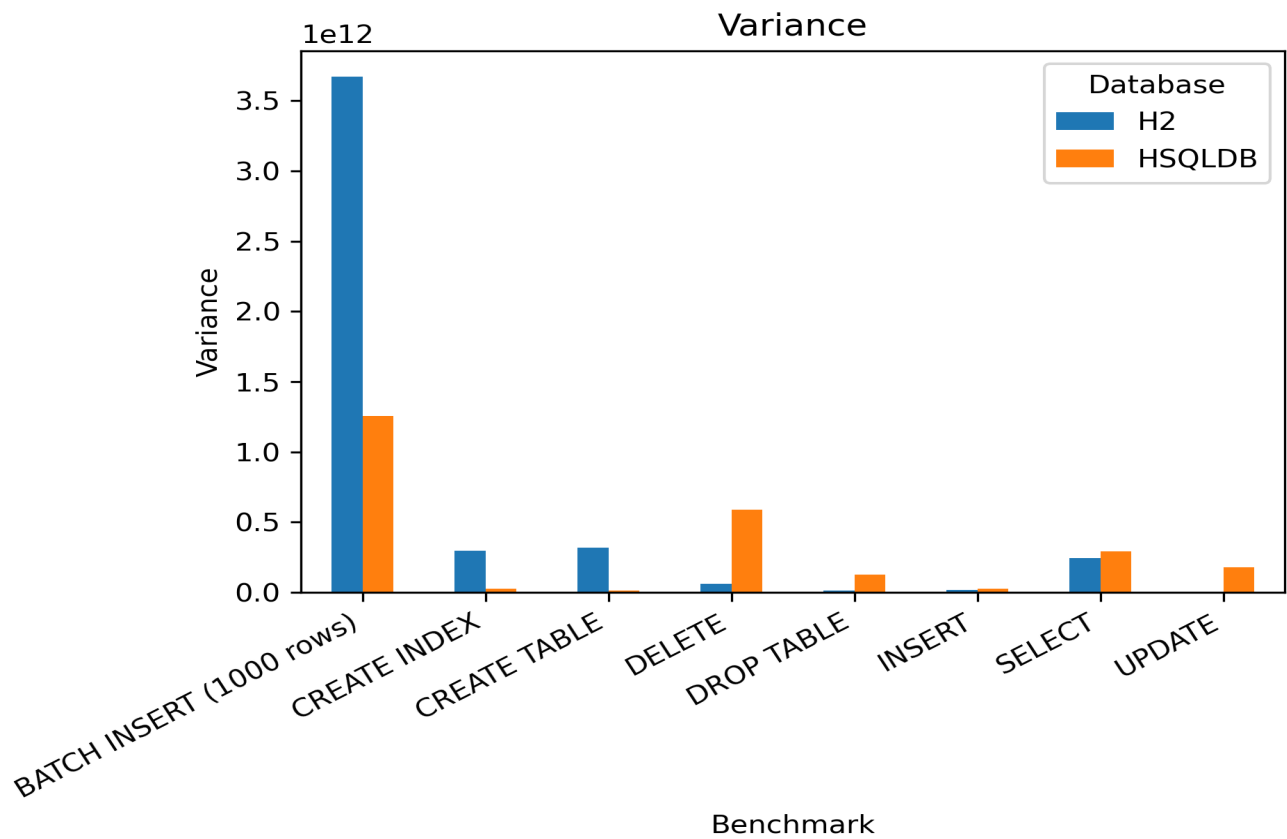
Median Execution Time



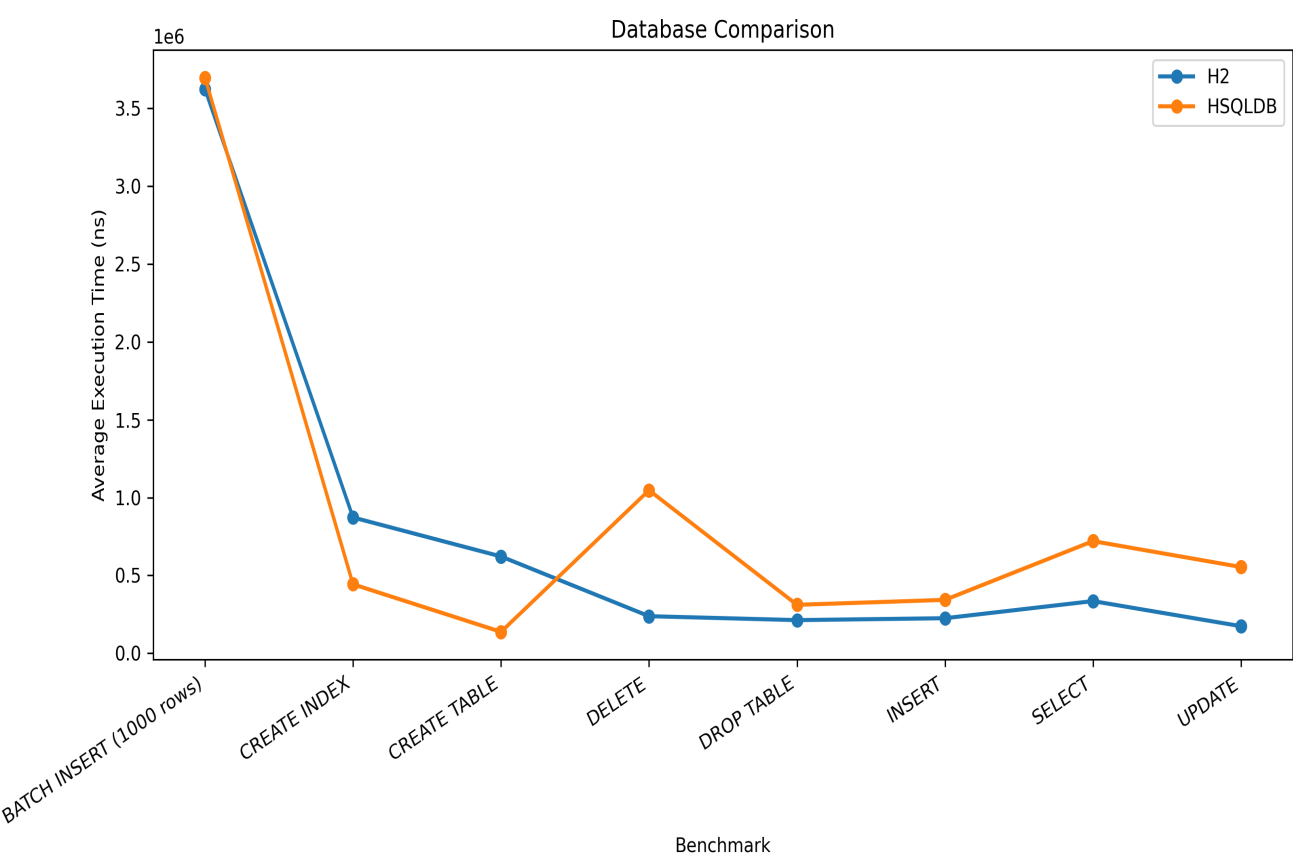
Standard Deviation



Variance



Database Comparison



Benchmark Winners

Benchmark	Winner	Average (ns)
BATCH INSERT (1000 rows)	H2	3,621,863
CREATE INDEX	HSQLDB	443,077
CREATE TABLE	HSQLDB	137,618
DELETE	H2	238,105
DROP TABLE	H2	212,929
INSERT	H2	225,121
SELECT	H2	335,075
UPDATE	H2	173,912

Overall Benchmark Score

Database	Benchmarks Won
H2	6
HSQLDB	2

Student's t-test

Benchmark	t-statistic	p-value	Significant?
BATCH INSERT (1000 rows)	-0.147	0.884271	NO
CREATE INDEX	3.309	0.003178	YES
CREATE TABLE	3.655	0.001491	YES
DELETE	-4.367	0.000228	YES
DROP TABLE	-1.146	0.263254	NO
INSERT	-2.488	0.017622	YES
SELECT	-2.289	0.027802	YES
UPDATE	-3.902	0.000933	YES

Interpretation: A p-value below 0.05 indicates that the observed performance difference between H2 and HSQLDB is statistically significant.

Automatic Conclusion

Summary

The benchmark suite executed 8 benchmark categories on 2 embedded database systems.

Based on the average execution time, **H2** achieved the best overall performance, winning 6 benchmark categories.

The fastest benchmark measured was **CREATE TABLE** executed on **HSQLDB** with an average execution time of **137,618 ns**.

The slowest benchmark measured was **BATCH INSERT (1000 rows)** executed on **HSQLDB** with an average execution time of **3,696,618 ns**.

Overall, the benchmark results indicate that execution times remained stable after warm-up iterations while demonstrating measurable performance differences between the tested database systems.

This report was generated automatically from the benchmark CSV produced by the Java benchmark framework.